

Childhood Inequities in Sedentarism

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Aim:

Examine trends in socioeconomic inequalities in leisure-time sedentarism among children aged 6 to 12 years in Spain between 2003 and 2023.

Methods:

Data source: Five cycles (2003, 2006, 2011, 2017, 2023) of the Spanish National Health Survey. Final study population of 19,500.

Exposure: The continuous socioeconomic rank variable derived from occupational social class of the household reference person.

Outcome: Children reported as engaging in any physical activity regardless of frequency were classified as non-sedentary, and those reported as performing none as sedentary.

Statistical Analysis: Quantification of socioeconomic inequalities in childhood sedentarism:

- All models adjusted for and sex, incorporating survey weights:
 - Poisson regression models with log link and robust standard errors, representing the Relative Index of Inequality - RII, interpreted as the prevalence ratio of sedentarism between the lowest and highest socioeconomic positions.
 - Poisson regression models with identity link, representing the Slope Index of Inequality - SII, interpreted as the absolute difference in predicted sedentarism prevalence, in percentage points, between the lowest and highest socioeconomic positions.
 - Multi-level Poisson regression models with a log link were fitted including all survey years. Individuals were nested within survey years and NUTS regions. Random intercepts and random slopes for social class were specified at the survey and region levels, allowing baseline sedentarism and socioeconomic inequalities to vary across time and regions.
 - * Intraclass correlation coefficients (ICCs) to identify regional clustering (random intercepts and slopes for social class).
- Annual percent change (APC) in sedentarism inequality overall and separately for each NUTS region from 2003 to 2023, where positive APC indicate increasing RII over time. Inflection point in the temporal trend of sedentarism inequality between 2003 and 2023 using segmented regression.

NUTS regions classification:

- The Canary Islands; Centre: Castile and Leon, Castilla-La Mancha, and Extremadura; East: Catalonia, the Valencian Community, and the Balearic Islands; Madrid: the Community of Madrid; North-East: the Basque Country, Navarre, La Rioja, and Aragon; North-West: Galicia, Asturias, and Cantabria; and South: Andalusia, Murcia, and the autonomous cities of Ceuta and Melilla.

Results:

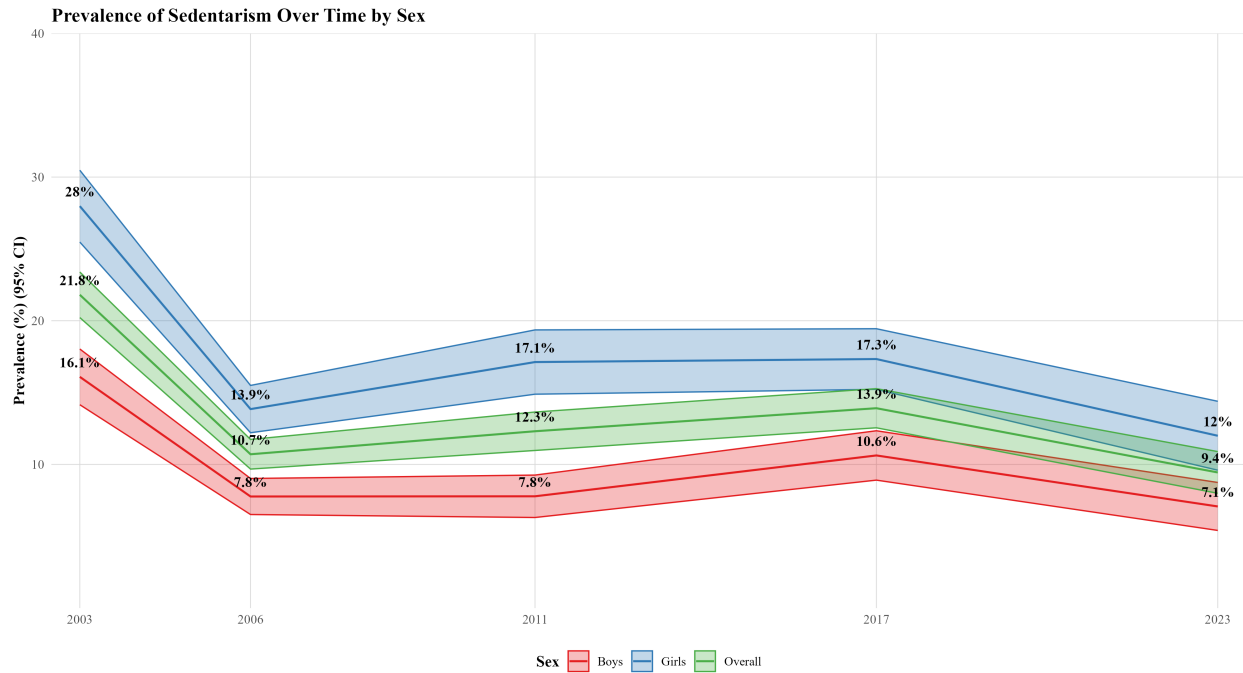


Figure 1: Prevalence of Sedentarism 2003-2023 by Sex

Sedentarism prevalence across survey years: Overall decrease, 21.8% (95% CI: 20.2, 23.4) in 2003 to 9.4% (95% CI: 8.0, 10.9) in 2023. Among girls, 12.0% (95% CI: 9.6, 14.4) and boys, 7.1% (95% CI: 5.4, 8.7) in 2023. Among the lowest socioeconomic position, 20.4% (95% CI: 14.8%, 26.0%), and among the highest, 5.8% (95% CI: 2.6, 9.0) in 2023.

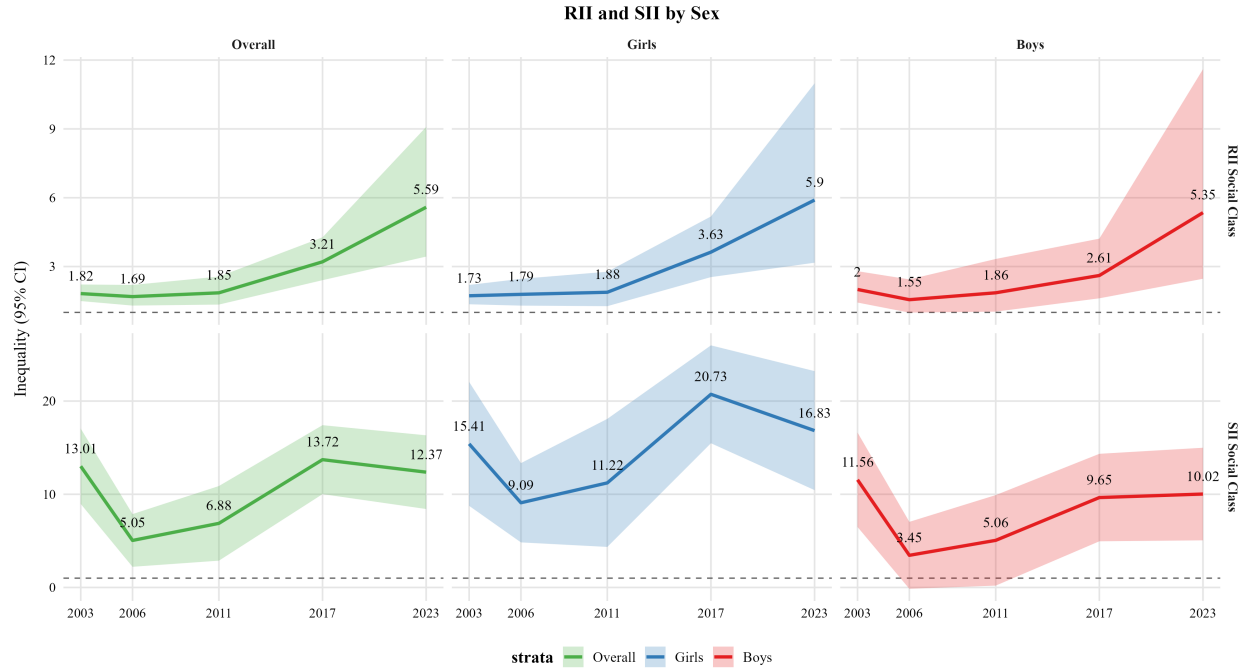


Figure 2: RII and SII overall and by sex

Inequalities in sedentarism across survey years: RII rose from 1.82 (95% CI: 1.50, 2.22) in 2003 to 5.59 (95% CI: 3.42, 9.08) in 2023, indicating that children in the lowest social class had more than five times the prevalence of sedentarism compared to those in the highest. In 2023, inequalities were slightly greater among girls at 5.90 (95% CI: 2.47, 11.50) than boys at 5.35 (95% CI: 3.17, 10.99).

SII was 13.01 (95% CI: 8.97, 17.05) percentage points higher among children in the lowest social class compared to the highest in 2003 and remained similar in 2023 at 12.37 (95% CI: 8.42, 16.33). Among girls, this absolute inequality reached 16.83 (95% CI: 10.45, 23.21) percentage points in 2023 and peaked at 20.73 (95% CI: 15.48, 25.98) percentage points in 2017.

Inequalities in Childhood Sedentarism by NUTS per Survey Year

Unit: Relative Index of Inequality

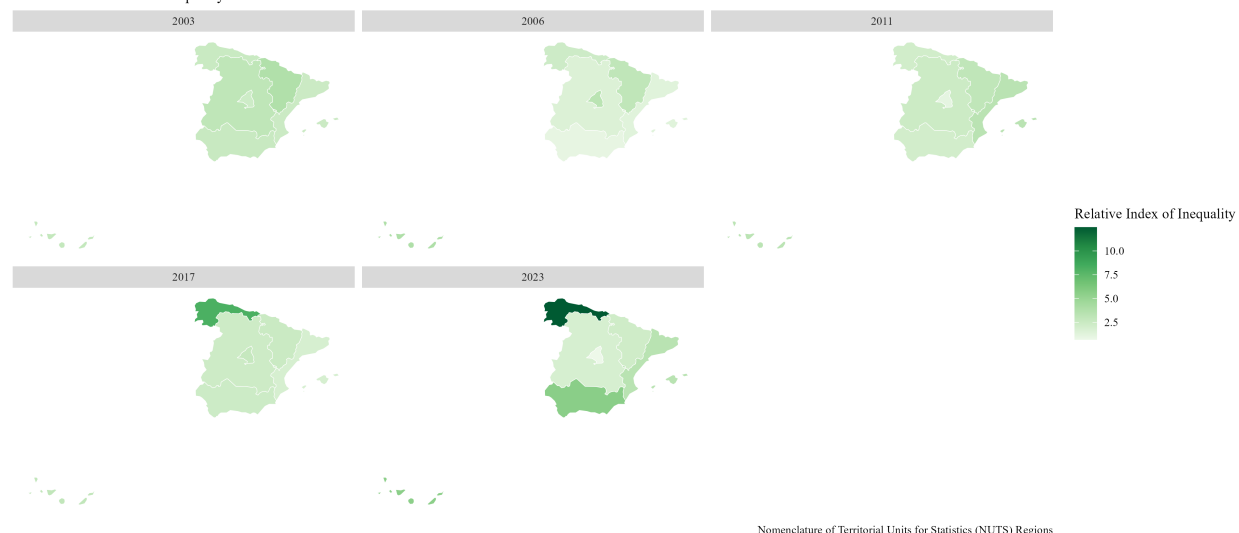


Figure 3: Overall: RII by NUTS per Survey Year

Inequalities in sedentarism across survey years and NUTS regions: In 2003, inequalities were present across all regions, with the highest observed in the North-East at 3.79 (95% CI: 2.58, 5.58). In contrast, Madrid experienced a marked decline in inequality over time, with the RII decreasing from 2.33 (95% CI: 1.58, 3.43) in 2003 to 0.69 (95% CI: 0.47, 1.02) in 2023, suggesting a reversal of the socioeconomic gradient. The North-West showed the greatest inequality overall, reaching an RII of 12.48 (95% CI: 8.49, 18.36) in 2023. Inequalities also increased over time in the Canary Islands at 5.58 (95% CI: 3.79, 8.20) and South regions at 5.65 (95% CI: 3.84, 8.31) percentage points.

Girls: Inequalities in Childhood Sedentarism by NUTS per Survey Year

Unit: Relative Index of Inequality

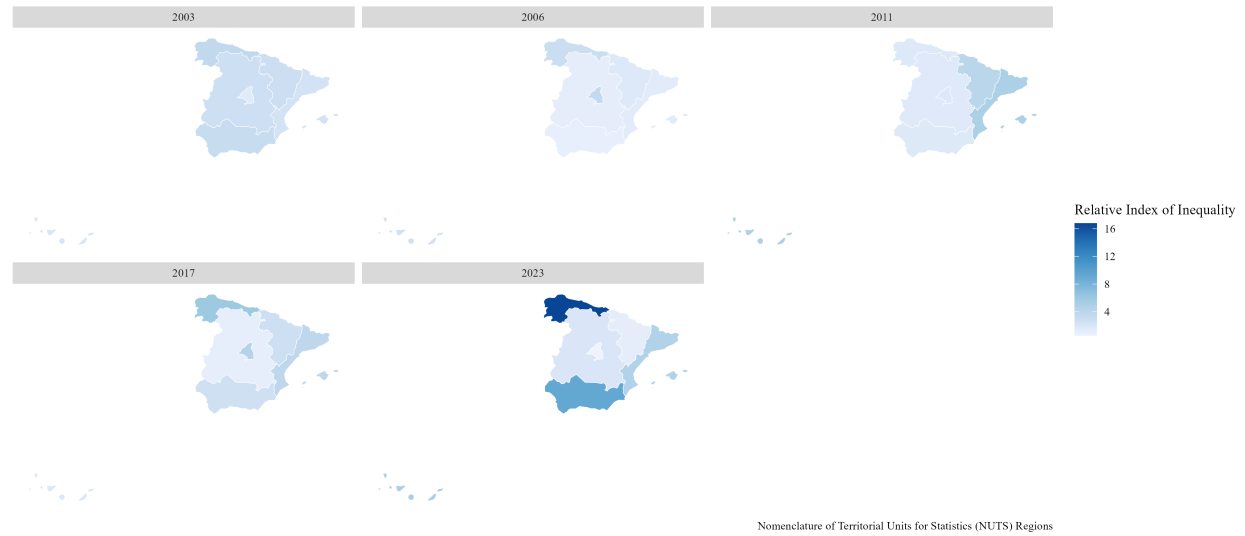


Figure 4: Girls: RII by NUTS per Survey Year

Girls - Inequalities in sedentarism across survey years and NUTS regions: Among girls, relative inequalities in sedentarism were present across all NUTS regions in 2003 and increased in several regions over time. The largest increase was observed in the North-West, with RII of 16.79 (95% CI: 11.58, 24.35) in 2023, followed by the South at 9.28 (95% CI: 6.40, 13.45), indicating substantially higher prevalence among girls in the lowest social class compared to the highest.

Boys: Inequalities in Childhood Sedentarism by NUTS per Survey Year

Unit: Relative Index of Inequality

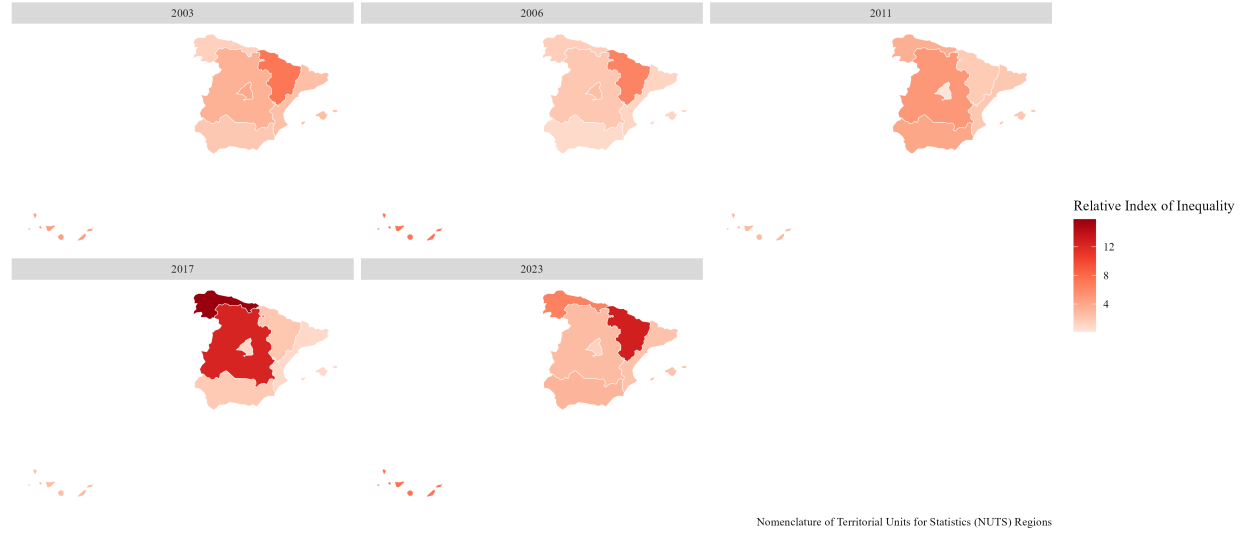


Figure 5: Boys: RII by NUTS per Survey Year

Boys - Inequalities in sedentarism across survey years and NUTS regions: In 2003, the highest inequality was observed in the North-East at 7.21 (95% CI: 4.16, 12.50). Inequalities increased in the Centre and North-West in 2017 with 12.48 (95% CI: 7.20, 21.64) and 15.85 (95% CI: 9.14, 27.48) percentage points, respectively, before declining in 2023. In contrast, the North-East showed a substantial increase by 2023, reaching 12.80 (95% CI: 7.38, 22.19).

ICCs could not be reliably estimated for models including random slopes for social class due to singularity issues. Using random-intercept models only, ICC values ranged from approximately 1% to 5%, indicating minimal regional clustering in sedentarism.

APCs and Inflection Point: From 2003 to 2023, the annual percent change in relative socioeconomic inequalities of sedentarism at the national level was 5.36% (95% CI: 2.57, 8.22). Among girls, the increase was slightly higher at 5.92% (95% CI: 2.39, 9.56), and among boys slightly lower at 4.64% (95% CI: 1.12, 8.29). Relative inequalities decreased in the North-East (-2.47, 95% CI: -3.12, -1.81) but increased substantially in the North-West (9.33, 95% CI: 4.85, 13.99). Among girls, inequalities increased over time in the Canary Islands (4.21, 95% CI: 2.33, 6.13) and East regions (3.34, 95% CI: 3.14, 3.54). While among boys, inequalities increased primarily in the North-West (9.57, 95% CI: 2.90, 16.68).

The segmented regression identified an approximate inflection point around 2014, situated between 2011 and 2017 survey years. Prior to this year, the slope of the trend in RII was -0.009 per year, indicating a relatively stable or slightly decreasing inequalities. After this point, the slope increased sharply (0.36 per year).